FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

COMPUTER SCIENCE DEPARTMENT

(PROGRAMMING LANGUAGES)

TECHNICAL ASSESSMENT
FINAL PROJECT

<STUDENT NAME>
<SECTION>
<DATE>

I. OBJECTIVES

At the end of this project, students must be able to:

Cognitive

a.) integrate concepts from lexical, syntax, semantic analysis, data types, control flow, and object orientation into a unified programming language design.

Psychomotor

a.) design and implement a simple programming language or interpreter applying compiler design principles.

Affective

a.) appreciate the complexity and importance of programming language design in software development.

II. BACKGROUND INFORMATION

In order to accomplish this project, the student must have a clear understanding of the following topics:

· Module 1: Introduction to Programming Languages

· Module 2: Lexical and Syntax Analysis

· Module 3: Names, Scope, and Binding

· Module 4: Semantic Analysis

· Module 5: Control Flow

· Module 6: Data Types and Type Systems

· Module 7: Data Abstraction and Object Orientation

This project integrates all modules to design a mini programming language or interpreter.

III. LABORATORY PROCEDURE

PROJECT TITLE:

Design and Implementation of a Mini Programming Language

PHASE 1: Language Design (Module 1 Integration)

a.) Define the purpose of your programming language
b.) Specify:

Syntax style (keywords, symbols)
Basic structure of programs
PHASE 2: Lexical Analysis (Module 2)

a.) Define tokens:

Identifiers
Keywords
Operators
Literals

b.) Create a simple lexical analyzer that scans input code

PHASE 3: Syntax Analysis (Module 2)

a.) Define grammar rules (basic structure)
b.) Validate program structure (correct syntax)

PHASE 4: Names, Scope, and Binding (Module 3)

a.) Implement variable declaration rules
b.) Define:

Scope (global/local)
Binding rules
PHASE 5: Semantic Analysis (Module 4)

a.) Perform:

Type checking
Variable declaration checking

b.) Detect semantic errors

PHASE 6: Control Flow Implementation (Module 5)

a.) Implement:

Conditional statements (if/else)
Loops (for/while)

b.) Demonstrate execution flow

PHASE 7: Data Types (Module 6)

a.) Support basic data types:

Integer
Float
Boolean
String

b.) Handle type conversion and operations

PHASE 8: Object-Oriented Features (Module 7)

a.) Implement:

Class definition
Object creation
Encapsulation (basic)

b.) Demonstrate usage

IV. QUESTION AND ANSWER

What is the purpose of your designed programming language?

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____
____

How does your lexical analyzer identify tokens?

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____
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What grammar rules did you define for syntax analysis?

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How does your system handle scope and binding?

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How does your system handle scope and binding?

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What semantic checks are implemented in your language?

What control flow structures are supported?.

What data types are supported and how are they used?

PROGRAMMING TASK

1. Mini Programming Language Implementation

Develop a program that:

a.) Accepts input code (string or file)

b.) Performs:

- Lexical Analysis
- Syntax Checking
- Semantic Checking

c.) Executes basic operations